CHAPTER 03 / TRAINING

Master the gym

Confidence comes one repetition at a time.

Learn the equipment, the movement and the intent.

How to use this guide

Read the explanations, try the practical examples and complete the worksheet. All training logs and menus are illustrative. Use the website for interactive tools; this document remains useful offline.

01 Understand

02 Put into practice

03 Review

Scope and safety

General education for adults, not a medical diagnosis, personalized treatment or a guaranteed transformation. Stop exercise for sharp pain or unusual symptoms. A qualified clinician, dietitian or coach can help when individual circumstances require assessment.

What you will learn

01. Your first session
02. Choose your equipment
03. Sets, repetitions and rest
04. Technique and breathing
05. RPE and RIR

01 / Your first session

Ask staff how to adjust machines and use rack safeties. Start with a short easy warm-up, then lighter practice sets of your first lifts. A warm-up prepares you; it should not exhaust you before training. Record seat positions for machines you will use again.

PUT IT INTO PRACTICE: Begin with a load that lets you learn the entire movement comfortably.

02 / Choose your equipment

Dumbbells let each side move independently. Barbells accommodate larger loads but may require a rack or spotter. Cables provide adjustable lines of pull. Machines guide the movement. A Smith machine follows a fixed rail; it is not identical to a free barbell. Bodyweight and bands are useful too.

PUT IT INTO PRACTICE: Choose equipment that matches your skill, comfort and available support.

03 / Sets, repetitions and rest

A repetition is one complete movement. A set is a group of repetitions before resting. Rest long enough to repeat the next set with control; a practical starting point is two to three minutes for demanding compound lifts and one to two for smaller movements. Adjust when breathing or performance has not recovered.

PUT IT INTO PRACTICE: Log 3 × 8 as three sets of eight repetitions, not 24 continuous reps.

04 / Technique and breathing

Move with control through a tolerable range. Keep joints supported and avoid bouncing or jerking the load. For general beginner lifting, breathe continuously and usually exhale through the harder phase. Bracing is a skill; prolonged breath-holding is not a requirement for beginners.

PUT IT INTO PRACTICE: Stop for sharp pain, dizziness or unusual shortness of breath and seek appropriate help.

05 / RPE and RIR

Resistance-training RPE describes set difficulty. An RPE of 10 is roughly no good repetitions remaining; RPE 8 is roughly two remaining. These are subjective approximations. Use them with load and repetition records, not as precise measurements.

PUT IT INTO PRACTICE: Choose an exercise below to review setup, execution, breathing and alternatives.

Practical reference

Exercise technique handbook

Goblet squat

Primary: Quadriceps | Secondary: Glutes, trunk | Equipment: Dumbbell

Setup	Hold one dumbbell at your chest; stand comfortably with feet near shoulder-width.
Execution	Sit between your hips as knees follow the toes; lower within a comfortable range; push through the floor to stand.
Breathing	Inhale during the descent and exhale as you stand.
Common mistakes	Heels lifting; knees collapsing inward; rushing the bottom.
Safety	Use a manageable load and stop if a joint hurts.
Alternative	Box squat

Romanian deadlift

Primary: Hamstrings | Secondary: Glutes, back stabilizers | Equipment: Dumbbells

Setup	Stand tall with weights close to the thighs and knees softly bent.
Execution	Move hips backward with weights near your legs; stop before your back position changes; stand by extending the hips.
Breathing	Breathe in and brace before hinging; exhale returning upright.
Common mistakes	Squatting instead of hinging; reaching too low; rounding under load.
Safety	Begin light; depth follows your controllable hip range.
Alternative	Cable pull-through

Push-up

Primary: Chest | Secondary: Triceps, shoulders, trunk | Equipment: Bodyweight

Setup	Place hands slightly wider than shoulders; keep a straight trunk.
Execution	Lower chest between hands with elbows at a comfortable angle; press away from the floor without sagging.
Breathing	Inhale down; exhale up.
Common mistakes	Dropping hips; flaring elbows excessively; incomplete control.
Safety	Use a raised surface if the floor version is too demanding.
Alternative	Incline push-up

Dumbbell bench press

Primary: Chest | Secondary: Triceps, anterior deltoids | Equipment: Bench and dumbbells

Setup	Lie on a stable bench with feet planted; hold weights over the chest.
Execution	Lower with forearms near vertical; press upward smoothly without banging weights together.
Breathing	Inhale lowering; exhale pressing.
Common mistakes	Bouncing; wrists folding; uncontrolled deep stretch.
Safety	Learn a safe setup and exit; use help for challenging loads.
Alternative	Machine chest press

Seated cable row

Primary: Back | Secondary: Biceps, rear shoulders | Equipment: Cable machine

Setup	Sit with feet braced and torso comfortably upright.
Execution	Pull handles toward the lower ribs; move the shoulder blades naturally; return slowly without throwing the torso.
Breathing	Exhale pulling; inhale returning.
Common mistakes	Yanking with the back; shrugging throughout; swinging.
Safety	Check the attachment and select a manageable stack weight.
Alternative	Chest-supported dumbbell row

Lat pulldown

Primary: Back | Secondary: Biceps | Equipment: Cable pulldown

Setup	Set the thigh pad; grip the bar comfortably wider than the shoulders.
Execution	Pull toward the upper chest as elbows move down; return to a comfortable overhead reach.
Breathing	Exhale pulling; inhale returning.
Common mistakes	Pulling behind the neck; excessive lean; swinging.
Safety	Avoid forcing painful shoulder positions.
Alternative	Assisted pull-up

Dumbbell shoulder press

Primary: Shoulders | Secondary: Triceps | Equipment: Dumbbells

Setup	Sit supported or stand stable; hold weights near shoulders.
Execution	Press upward through a comfortable path; lower slowly to your starting position.

Setup	Sit supported or stand stable; hold weights near shoulders.
Breathing	Exhale pressing; inhale lowering.
Common mistakes	Overarching the lower back; forcing range; uneven control.
Safety	Choose a pain-free grip and range.
Alternative	Landmine press

Lateral raise

Primary: Shoulders | Secondary: Upper trapezius | Equipment: Dumbbells

Setup	Stand with light weights at your sides and elbows softly bent.
Execution	Raise arms out and slightly forward to a comfortable height; lower under control.
Breathing	Exhale lifting; inhale lowering.
Common mistakes	Swinging heavy weights; sharply shrugging; locking elbows.
Safety	Start light and avoid a painful shoulder arc.
Alternative	Cable lateral raise

Dumbbell curl

Primary: Biceps | Secondary: Forearms | Equipment: Dumbbells

Setup	Stand tall with weights at sides and upper arms near torso.
Execution	Bend the elbows without swinging your trunk; lower until arms comfortably straighten.
Breathing	Exhale curling; inhale lowering.
Common mistakes	Leaning back; swinging; bending wrists.
Safety	Use controlled loads and a comfortable wrist position.
Alternative	Cable curl

Triceps pushdown

Primary: Triceps | Secondary: Forearm stabilizers | Equipment: Cable machine

Setup	Stand at a high pulley with elbows near the sides.
Execution	Straighten the elbows without driving shoulders forward; return slowly.
Breathing	Exhale extending; inhale returning.
Common mistakes	Torso swinging; elbows moving excessively; wrists collapsing.
Safety	Check attachment security and avoid forced elbow lockout.
Alternative	Band pushdown

Split squat

Primary: Quadriceps | Secondary: Glutes, trunk | Equipment: Bodyweight or dumbbells

Setup	Stand in a staggered stance wide enough for balance.
Execution	Lower straight down with the front foot planted; stand using the front leg while the rear leg assists.
Breathing	Inhale down; exhale up.
Common mistakes	Narrow unstable stance; bouncing; rushing.
Safety	Use support for balance and begin without added load.
Alternative	Reverse lunge

Hip thrust

Primary: Glutes | Secondary: Hamstrings, trunk | Equipment: Stable bench and padded weight

Setup	Rest upper back against a secure bench, feet planted and knees bent.
Execution	Raise hips to a comfortable straight trunk position; pause briefly; lower under control.
Breathing	Exhale lifting; inhale lowering.
Common mistakes	Overarching the back; unstable bench; feet slipping.
Safety	Pad the load and secure the bench; start with bodyweight.
Alternative	Glute bridge

Leg curl

Primary: Hamstrings | Secondary: Calves | Equipment: Leg curl machine

Setup	Adjust the machine axis to your knee and pads as directed.
Execution	Bend the knees smoothly; return without letting the weight stack slam.
Breathing	Exhale curling; inhale extending.
Common mistakes	Lifting hips; swinging; incorrect knee alignment.
Safety	Ask staff to verify the machine setup.
Alternative	Slider leg curl

Leg extension

Primary: Quadriceps | Secondary: Minimal assistance | Equipment: Leg extension machine

Setup	Align knees with the machine pivot and position the shin pad comfortably.
Execution	Straighten knees under control; return gradually within a comfortable range.
Breathing	Exhale extending; inhale returning.

Setup	Align knees with the machine pivot and position the shin pad comfortably.
Common mistakes	Kicking the stack; poor alignment; forcing discomfort.
Safety	Adjust or stop if knee pain occurs.
Alternative	Supported squat

Calf raise

Primary: Calves | Secondary: Foot stabilizers | Equipment: Bodyweight or machine

Setup	Stand with support and feet comfortably aligned.
Execution	Lift heels, pause briefly and lower smoothly; keep the movement at the ankle.
Breathing	Exhale lifting; inhale lowering.
Common mistakes	Bouncing; rolling ankles; rushing the bottom.
Safety	Hold a support and start on flat ground.
Alternative	Seated calf raise

Dead bug

Primary: Abdominals | Secondary: Hip flexors | Equipment: Mat

Setup	Lie on your back with arms up and hips and knees bent.
Execution	Slowly reach an opposite arm and leg away while keeping the trunk controlled; return and alternate.
Breathing	Exhale as limbs extend; inhale returning.
Common mistakes	Arching the lower back; moving too fast.
Safety	Shorten the reach if trunk control is lost.
Alternative	Heel tap

Plank

Primary: Abdominals | Secondary: Glutes, shoulder stabilizers | Equipment: Mat

Setup	Support yourself on forearms and toes, elbows under shoulders.
Execution	Hold a steady trunk while breathing; end the set before the position collapses.
Breathing	Breathe steadily throughout.
Common mistakes	Holding the breath; sagging; raising hips excessively.
Safety	Use knees or an incline if needed; stop for shoulder pain.
Alternative	Incline plank

Farmer carry

Primary: Forearms | Secondary: Trunk, upper back | Equipment: Dumbbells

Setup	Stand tall holding manageable weights with a secure grip.
Execution	Walk slowly with controlled steps and upright posture; set weights down safely.
Breathing	Breathe continuously.
Common mistakes	Leaning; rushing; choosing weights you cannot set down.
Safety	Clear the walking path and avoid carrying to grip failure.
Alternative	Static dumbbell hold

Your study worksheet

Knowledge check

What should you do before using an unfamiliar machine?

A. Copy the heaviest lifter

B. Ask how to adjust and use it

C. Skip the warm-up

Reflect and apply

Prompt	Your response
One idea I can explain in my own words	
One action I can try this week	
What I will observe or measure	
What would make me seek professional guidance	
My next review date	

Chapter summary

Learn setup, repeat controlled movements, rest adequately and record what you did.

Answer and explanation

B. Correct setup and a light practice set help you learn safely.

Sources & limitations

These sources inform the educational principles. Original examples and planning templates are not study findings. No institution listed below endorses this journal. References checked November 2026.

ACSM (2026) — Resistance training prescription: overview of reviews

https://pubmed.ncbi.nlm.nih.gov/41843416/

OpenStax — Anatomy and Physiology 2e

https://openstax.org/books/anatomy-and-physiology-2e/pages/11-introduction

Interpretation matters

Research describes groups under particular conditions. Individual responses vary. The website models are simplified illustrations, not precise physiological simulations. Food values and calorie estimates are approximate. Use consistent observation and individualized professional support when needed.